

Capital Budgeting and Infrastructure Project Valuation with Monte Carlo Risk Simulation (INR)

Project Domain: Clean Infrastructure & Energy Finance	Target Asset: Rajasthan 100MW Solar Park
Data Source: Kaggle / The World Bank PPI Database	Currency Standard: Indian Rupees (₹ Crores & ₹/kWh)
Primary Methodology: Multi-Stage DCF & WACC Hurdle Rate	Risk Simulation: 5,000-Iteration Stochastic Monte Carlo

1. Executive Overview & Real-World Motivation (INR)

Large-scale clean infrastructure assets—such as a **100MW Solar PV Power Plant in Rajasthan (₹250 Crores)** or a 50MW Wind Farm in Tamil Nadu (₹350 Crores)—require substantial upfront capital expenditure and operate over 20-to-25-year Power Purchase Agreements (PPAs). Institutional lenders and CFOs require rigorous quantitative models to evaluate debt repayment and tariff viability under macroeconomic shocks.

2. Public Dataset Architecture & Attribute Dictionary (INR)

Column Name	Financial Definition	Baseline Value (INR)	Role in Valuation
Project_Name	Asset name and technology type	100MW Solar Park	Identifier
CapEx_INR_Crores	Initial construction and land cost	₹ 250.0 Crores	Year 0 cash outflow (\$I_0\$)
Annual_Gen_MWh	Total annual electricity generated	180,000 MWh	Volume driver for revenue
Base_Tariff_INR_kWh	Contracted PPA tariff per unit	₹ 3.85 / kWh	Price driver for revenue
Lifetime_Years	Asset operational lifecycle	20 Years	Discounting horizon (\$N\$)
Annual_OpEx_Crores	Maintenance, labor, grid fees	₹ 12.5 Crores/yr	Operating cash outflow
WACC_Percent	Weighted Average Cost of Capital	8.00%	Hurdle rate / discount rate
Degradation_Pct	Annual efficiency loss of panels	0.50% / year	Annual generation decay

3. Core Financial Engineering Mathematical Formulations (INR)

$$\text{Revenue}_t \text{ (₹ Cr)} = [\text{Generation (MWh)} \times 1,000 \times (1 - \text{Degradation})^{t-1} \times \text{Tariff (₹/kWh)}] / 10,000,000$$
$$\text{OpEx}_t \text{ (₹ Cr)} = \text{Base OpEx} \times (1 + \text{Inflation})^{t-1}$$
$$\text{Net Cash Flow (CF}_t\text{)} = (\text{Revenue}_t - \text{OpEx}_t) \times (1 - \text{Tax Rate})$$
$$\text{NPV (₹ Cr)} = -\text{CapEx} + \sum_{t=1}^N [\text{CF}_t / (1 + \text{WACC})^t]$$
$$\text{LCOE (₹/kWh)} = [\text{Total Discounted Costs (₹)}] / [\text{Total Discounted Energy (kWh)}]$$

4. Baseline Results Scorecard (Rajasthan 100MW Solar Park - ₹ INR)

Financial Performance Metric	Calculated Value (INR)	Target Hurdle	Financial Interpretation
Net Present Value (NPV)	+₹ 275.50 Crores	> ₹ 0.00 Cr	Highly value-accretive; project viable
Internal Rate of Return (IRR)	18.52%	WACC = 8.00%	Generates +10.52% excess return spread
Levelized Cost of Energy (LCOE)	₹ 1.99 / kWh	Tariff = ₹ 3.85 / kWh	Robust profit margin of +₹ 1.86 / kWh
Profitability Index (PI)	2.10x	> 1.00x	Every ₹1.00 invested yields ₹2.10 in PV
Discounted Payback Period	7 Years	< 20 Years	Capital fully recovered in Year 7
Probability of Loss	0.00% (Low Risk)	< 15.00%	Resilient under 5,000 market shocks
95% Value-at-Risk (VaR)	+₹ 152.80 Crores	> ₹ 0.00 Cr	Profitable even in worst 5% scenario